

ARI: An Autonomous Research Infrastructure

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Abstract

We describe **ARI**, an autonomous research infrastructure that couples a tree-search-based exploration loop with a paper-generation pipeline. The exploration loop is a best-first tree search (BFTS) over research-step nodes; expansion uses a Reason-and-Act (ReAct [1])-style agent that issues tool calls into a registry of fourteen domain skills. The paper pipeline turns the surviving lineage into a draft, then iteratively refines it under a hierarchical rubric. Every artefact lives inside a single checkpoint directory, which is the unit of reproducibility, lineage, and cost accounting. This report formalises the algorithms and the design rationale, reports preliminary observations, and discusses the determinism / novelty trade-off that constrains the design.

1 Introduction

1.1 Motivation

Research workflows in machine learning and adjacent fields share a common skeleton: brainstorm a research idea, design and run an experiment, analyse the result, write a paper, and iterate. Each step is increasingly tractable for large language models, but stitching them into a single autonomous loop has, so far, been an open engineering problem. The published systems that come closest — AIDE, the AI Scientist family, VirSci, and PaperBench evaluators — each fix one or two of these axes while leaving the others manual.

ARI exists to fill that gap. We assume the user supplies a research question and a compute budget; the system returns a publishable paper, the experimental code that backs every claim, and a complete audit trail of how it got there. The audit trail is non-negotiable: every node of the search tree, every tool call, every model invocation, and every dollar of API spend is recorded inside a single *checkpoint directory* ([2] adopt a similar discipline). The checkpoint is the unit of reproducibility.

1.2 Contributions

This report formalises three contributions:

1. A **best-first tree search** over research-step nodes whose selection score $U(n)$ separates the empirical reward, a novelty term, and a depth penalty (section 3.1).
2. A **paper-generation pipeline** keyed off a hierarchical rubric R whose leaf judges grade individual claims; the system score is $\text{score}(P) = \sum_i w_i \text{judge}_i(P)$ (section 4.2). The PaperBench replicator is integrated as a tool call (section 4.4).
3. A **checkpoint scoping** discipline that makes the pipeline fully reproducible: every external

state (memory, cost ledger, lineage pointers) is written below `$ARI_CHECKPOINT_DIR`, never the user's home directory. This is the operational consequence of the *determinism* design principle, one of five that section 2.5 lays out in full (we refer to it as P2 there and in the rest of the report).

1.3 Scope of this report

This report describes **ari-core** and the fourteen **ari-skill-*** packages at version v0.8.0, which promotes the PaperBench reproduction path to a protocol-faithful three-stage contract — agent rollout, reproduction, and grading — adds host-truthful environment guardrails that stop the rollout agent from assuming capabilities the host does not provide, makes the search-evaluation layers configurable, and carries out a first operational reproduction pass on a real lossy-compression target (section 5). The body describes the algorithms and data flow at the module level; the verbatim LLM prompts the system uses are listed in section B. The empirical study is preliminary at this version — section 5 reports the observations available so far, and a controlled benchmark is left to future work.

1.4 Organisation

Section 2 gives a top-down view of ARI: the orchestrator, the fourteen skills, the pipeline DAG, and the design principles. Sections 3 and 4 are the two technical cores, each with its own formal definitions and empirical observations. Section 5 reports preliminary observations; section 6 positions the work; section 7 collects open problems.

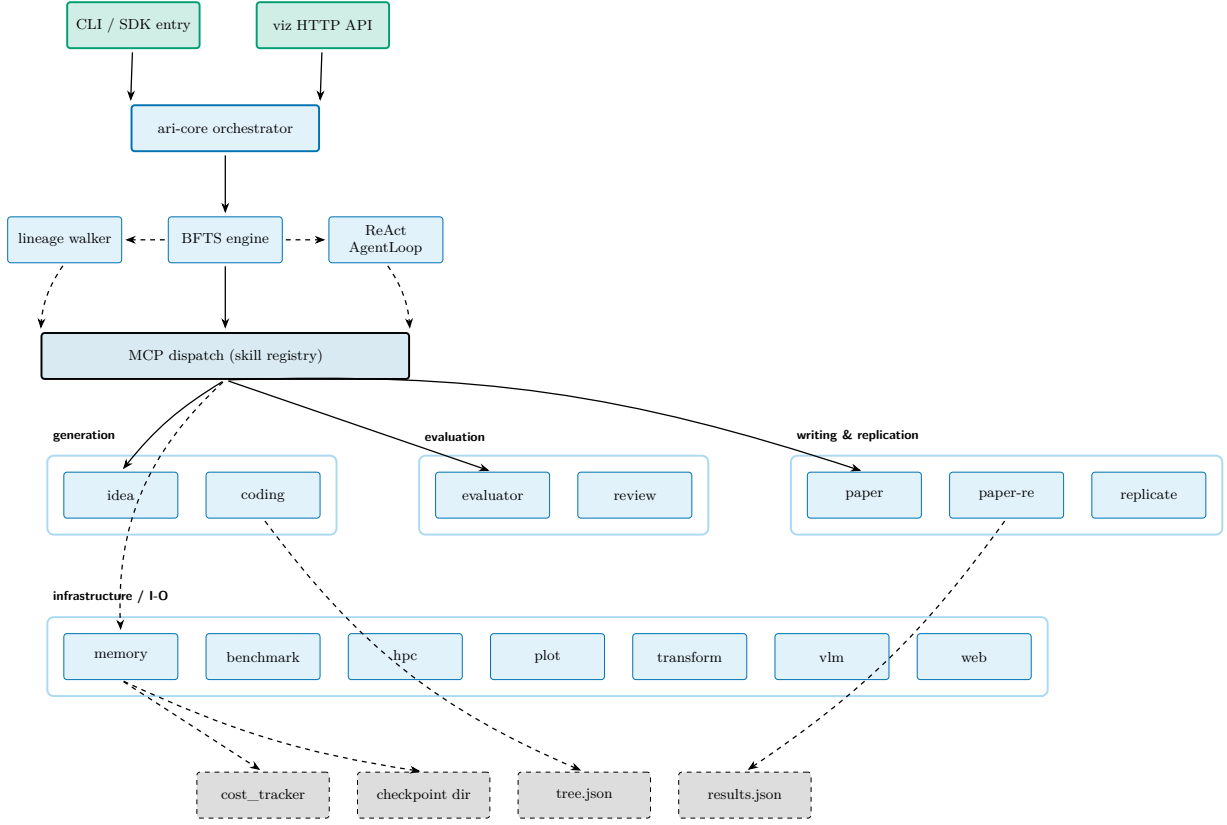


Figure 1: The full ARI stack at v0.8.0. The CLI / SDK entry-points drive a single orchestrator (**ari-core**); the orchestrator owns three engines (BFTS, lineage walker, ReAct **AgentLoop**) and dispatches into the 14-skill registry through the MCP layer. All persistence lands inside one checkpoint directory: cost ledger, **tree.json**, and **results.json**. Solid edges are control; dashed edges are state.

2 System Overview

2.1 The full stack

Figure 1 draws the full ARI stack at v0.8.0. The system splits into one driver and many tool providers. The driver (**ari-core**) owns the orchestrator, the BFTS engine, the lineage walker, the agent loop, the checkpoint, the cost tracker, and the configuration loader. Each skill is a separate Python package shipped as a *Model Context Protocol* (MCP) server with its own `skill.yaml` declaration. MCP is the JSON-RPC-over-stdio interface that Anthropic standardised for exposing tool-style capabilities to an LLM agent loop; ARI uses it as the universal seam between **ari-core** and every skill. The fourteen skills currently in tree are: **benchmark**, **coding**, **evaluator**, **hpc**, **idea**, **memory**, **orchestrator**, **paper**, **paper-re**, **plot**, **replicate**, **transform**, **vlm**, and **web**.

The orchestrator dispatches calls but does not perform domain work itself. This separation matters because skills can be added or swapped without touching the search algorithm: the BFTS engine is unaware of which skills participate; everything is mediated by the MCP dispatch layer.

The smaller diagram in fig. 2 below is the “operational” view (control + state outputs) used in the rest of the report; fig. 1 is the “layered” view (driver → MCP → skills → persistence).

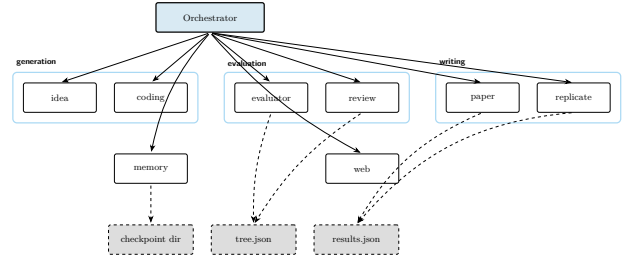


Figure 2: Operational view: orchestrator dispatches into the skill registry; every skill writes only into the active checkpoint directory. Dashed edges indicate state.

2.2 Pipeline DAG

A run reduces to a four-station pipeline (fig. 3). The idea station selects or generates research questions; the exploration station spawns nodes under BFTS; the paper station compiles the best lineage into a draft; and the review station grades the draft against the rubric. The graph is a DAG plus one back edge from review to paper: only this edge can introduce non-monotonicity (refining worsens an axis), and the *convergence criterion* (section 4.3) is what bounds the back-edge count.

The pipeline is driven by a single runner in the orchestrator; a scientific-data assembler closes the gap between exploration output and paper input, collating the per-node artefacts the writer needs.

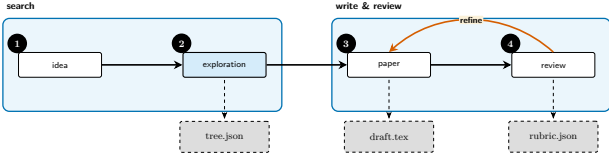


Figure 3: Pipeline DAG: idea $\rightarrow$ exploration $\rightarrow$ paper $\rightarrow$ review, with *refine* closing the back edge. Dashed outputs are persisted to the checkpoint; the refine cycle rewrites the draft and re-runs the review.

2.3 BFTS control flow

Figure 4 traces a single BFTS step end-to-end. The implementation is laid out in section 3; the diagram fixes the vocabulary used by the rest of the report (frontier, selection, fallback, expand, prune, stagnation) and the prompt files (`bfts_select.md`, `bfts_expand_select.md`, `bfts_expand.md`) that materialise each LLM-driven step.

2.4 paper-re component view

Figure 5 is the second-level view of *paper-re*. ARI’s contribution at this layer is small but specific: a thin adapter wraps each agent tool so its output is size-bounded and redirects the vendored solver’s hard-coded paths to ARI’s per-node workspace; everything else flows through PaperBench upstream verbatim, which keeps ARI aligned with the upstream replication-task framing. Section 4.4 develops this integration.

2.5 Checkpoint scoping and the design principles

ARI is built around five *design principles* (P1–P5). The binding one is **P2: determinism**. Every randomised step records its seed; every LLM call records its model identity, prompt, and output; every tool call is tagged with its node identity (a copy-on-write guard) so its writes land inside the right node’s working tree. Re-running a checkpoint must reproduce the same artefacts byte-for-byte, except for fields explicitly marked as wallclock or as model-side non-determinism.

The other principles, in summary:

- **P1 single artefact tree:** every output of a run lives below `$ARI_CHECKPOINT_DIR`. Nothing leaks to `$HOME` or `/tmp`.
- **P3 small components:** each skill is a separate package; replacing one skill does not require rebuilding *ari-core*.
- **P4 explicit lineage:** every node and every checkpoint record their parent, so a derivative experiment can recompute only its delta.
- **P5 cost transparency:** the cost tracker attaches dollar costs to every model call so the user can bound a run.

The checkpoint serialiser writes three top-level files: the full search tree to `tree.json`, per-node payloads to `nodes_tree.json`, and end-of-run aggregates to `results.json`. Lineage walks across checkpoints climb from a child to its ancestors.

Figure 6 draws the on-disk shape of a checkpoint: shared run-level files on the left, per-node working trees on the right. The shape is the contract: any tool that takes ARI’s output as input (a follow-up run, a regression check, a paper draft) points at a single such directory.

3 Exploration Algorithm

3.1 Formalisation

The exploration loop maintains a tree $\mathcal{T} = (\mathcal{N}, E)$ in which each node $n \in \mathcal{N}$ represents a single research step: an idea revision, a code change, a benchmark run, an analysis, etc. A node carries

- a discrete state $s(n) \in \{\text{open}, \text{eval}, \text{done}, \text{failed}\}$,
- a categorical *label* drawn from the `NodeLabel` enumeration,
- per-axis evaluation metrics in a freeform dict `node.metrics`, including a distinguished scalar `_scientific_score` $\in [0, 1]$,
- a Boolean `has_real_data` that flags whether the node carries actual measurements rather than provisional text, and
- the bookkeeping field `depth` surfaced both in the selection prompts and to the pruning predicate. ARI does not retry failed nodes, so no `retry_count` signal is surfaced.

The aggregation of axis-level signals into the scientific score is delegated to any implementation of the *Evaluator* protocol; ARI does not pin a fixed linear combination. The protocol is the only seam between BFTS and downstream rubric evaluation, which keeps the search engine agnostic about how axes get reduced to a scalar.

Run-loop state. Each iteration of the outer loop transforms the tuple

$$S = (\mathcal{F}, \mathcal{P}, e, H),$$

where $\mathcal{F} \subseteq \mathcal{N}$ holds completed nodes (done / failed) eligible for expansion, $\mathcal{P} \subseteq \mathcal{N}$ holds nodes in state open ready to execute, $e : \mathcal{N} \rightarrow \mathbb{N}$ counts how many times each frontier node has been expanded, and $H = (l_1, \dots, l_t)$ is the sliding window of recently-executed labels (length capped at $W = 20$, see section 3.3). The initial state is $S_0 = (\emptyset, \{n_{\text{root}}\}, \mathbf{0}, ()$.

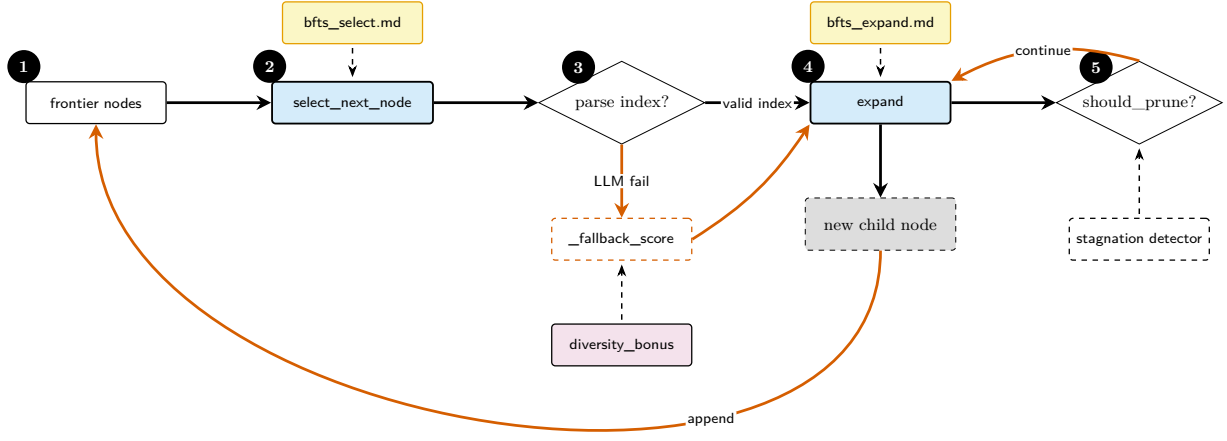


Figure 4: BFTS control flow. Two LLM-driven decision points (`select_next_node`, `expand`) each consult a separate prompt file; the selection decision falls back to a deterministic `_fallback_score` (which combines `_scientific_score` with `diversity_bonus`) when the LLM returns an unparseable index. Pruning is gated by both `should_prune` (max-total-nodes hard cut) and the stagnation detector. Detail in section 3.

Pruning predicate. The hard exploration budget is the predicate

$$\Pi(n; |\mathcal{N}|) = \mathbb{K}[|\mathcal{N}| \geq B_N] \vee \mathbb{K}[\text{depth}(n) \geq B_D] \vee \sigma(n), \quad (1)$$

where $B_N = \text{config.max_total_nodes}$, $B_D = \text{config.max_depth}$, and the *sterile* predicate

$$\sigma(n) = \mathbb{K}[|\Delta_n| = 0] \quad (2)$$

(with Δ_n the set of files added, modified, or deleted by n) fires when a node finishes its agent loop without writing any new artefact relative to its parent. The run loop computes this file diff and records the result as a Boolean `_sterile` flag on the node; `should_prune` reads that flag together with the two caps. The caller passes the live $|\mathcal{N}|$ each iteration so there is exactly one source of truth for the node count. Step / cost budgets are enforced separately by the cost tracker at the call site, not inside the BFTS engine.

Retire predicate. On top of Π , the run-loop applies a softer *frontier retire predicate* ρ to drop a parent f from $\mathcal{F}$ once it is no longer the most productive lead:

$$\rho(f, c) = \underbrace{\mathbb{K}[r(c) > r(f)]}_{\text{Rule A}} \vee \underbrace{\mathbb{K}[e(f) \geq E_{\max}]}_{\text{Rule B}}, \quad (3)$$

with $E_{\max} = \text{config.max_expansions_per_node}$ (default 4). Rule A retires f as soon as a child c outscores it; Rule B caps the per-parent budget so the search spreads. ρ does *not* delete the node — its history stays on $\mathcal{T}$ — only its eligibility to be re-expanded is revoked.

The full step decomposition of one outer iteration (inner expansion sub-loop, parallel ReAct batch, post-execution retire) is shown in fig. 7; the explicit pseudocode is algorithm 1 in section 3.2.

3.2 Run-loop algorithm

Algorithm 1 writes the iteration as explicit pseudocode. The inner WHILE fills empty worker slots one expansion at a time (so workers always start running before the next expansion is proposed); the outer block then assembles a batch by calling `select_next_node` repeatedly — one node per call — until the batch reaches the worker cap, and executes that batch in parallel; the post-execution block (a) records the executed label into H so the diversity bonus reflects what actually ran, and (b) applies ρ Rules A and B.

3.3 Node selection (LLM-driven)

ARI deliberately does *not* reduce node ranking to a closed-form scalar utility. Instead, both `select_next_node` (“which pending node should the next agent step run?”) and `select_best_to_expand` (“which completed node is most worth expanding?”) hand a list of candidate descriptions to the LLM and parse back a single 0-based index — each call selects exactly one node. The candidate description for each node n is a one-liner of the form

```
[i] id=..., depth=..., label=...,
has_real_data=..., metrics={...},
summary=...
```

The run-selection variant (`select_next_node`) additionally appends `diversity_bonus+=0.05` to the line for any node that currently earns the bonus; `select_best_to_expand` omits it. The prompts that consume the list are reproduced verbatim as section B.3 (selection) and section B.2 (expansion target). The selection criteria the prompt enumerates are: nodes with `has_real_data=True` and strong metrics are high-value; metrics are weighed holistically (multi-objective); deeper nodes with excellent results

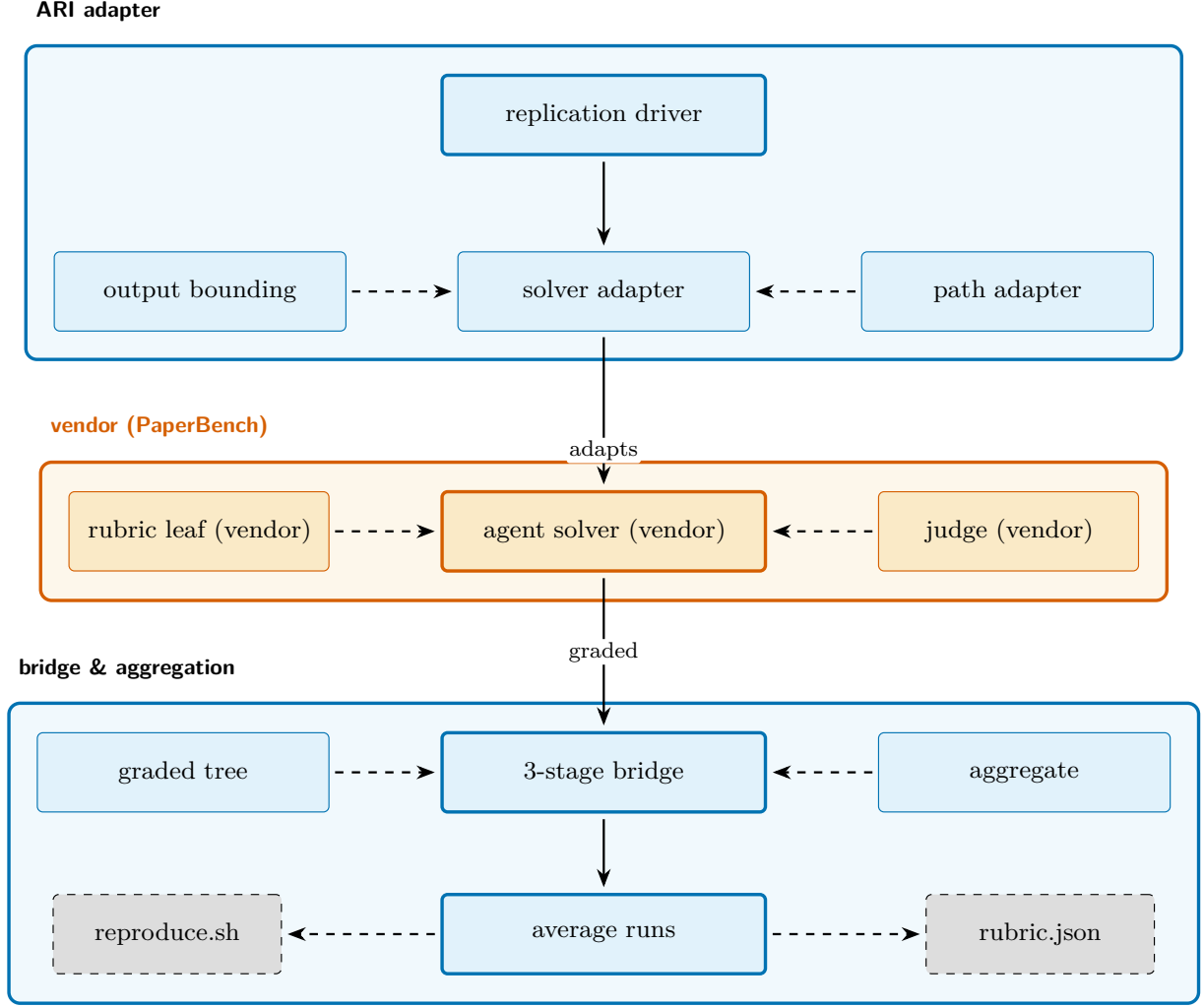


Figure 5: The paper-re skill’s components in three layers. The top row is ARI’s adapter layer (replication driver, solver adapter, output bounding, path adapter); the middle row is the vendored PaperBench package (agent solver, rubric leaves, judge); the bottom row is the bridge that collapses a per-submission graded tree to a single `rubric.json` score plus an ARI-side `reproduce.sh`. Detail in section 4.4.

are worth continuing; the `diversity_bonus` is treated as a soft tiebreaker only. The `label` field matches the information shown to `select_best_to_expand`, and there is no spurious “low retry” criterion.

Diversity bonus

The diversity bonus `BFTS.diversity_bonus` tracks recent labels in `_recent_label_history` (a sliding window populated by `record_run`). For a node whose label is *under-*represented relative to the most common label in that window, the function returns

$$\text{div}(n) = \begin{cases} +0.05 & \text{cnt}(n) \leq \frac{1}{2} \text{cnt}_{\max}, \\ 0 & \text{otherwise,} \end{cases} \quad (4)$$

where $\text{cnt}(n)$ is the number of times $\text{label}(n)$ occurs in the sliding window and $\text{cnt}_{\max} = \max_{\ell} \text{cnt}(\ell)$. The constant 0.05 is small by design: scientific score still dominates ranking, and the bonus only breaks near-ties.

LLM fallback

If the LLM returns an unparseable index, `select_next_node` falls back to a deterministic ranking via the `_select_fallback` helper (which ranks candidates by the `_fallback_score` expression). The default scoring expression is

$$\text{fb}(n) = r(n) + \text{div}(n), \quad (5)$$

where $r(n) \equiv \text{_scientific_score}(n)$ is the node’s scalar reward (section A). `_select_fallback` prefers nodes with `has_real_data=True` when any are available, then returns the candidate with the highest $\text{fb}(\cdot)$. This guarantees the loop always makes progress even when the LLM call degrades.

Configurable evaluation layers

The four evaluation surfaces in sections 3.1 and 3.3 are independently configurable; the defaults reproduce the equations above, so an unmodified configuration is a no-op.

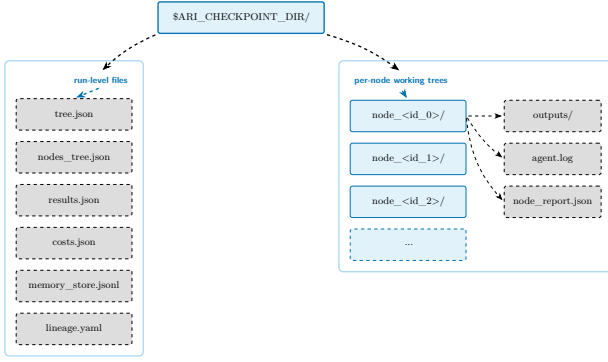


Figure 6: Checkpoint directory layout. Run-level files (left band) are written by the orchestrator; per-node working trees (right band) hold the artefacts produced inside one BFTS expansion. Reproducing a run is exactly “re-execute against this directory”, and the layout guarantees that no other state is needed.

Composite formula. Selected by `evaluator.composite`. The reduction $\{a_k\} \rightarrow \text{_scientific_score}$ is one of: weighted harmonic mean (default), arithmetic mean, bottleneck `weighted_min`, or weighted geometric mean. The harmonic and geometric forms apply a floor $\epsilon = 0.01$ to a zero-valued axis so the denominator stays finite.

Frontier-score strategy. Selected by `bfts.frontier_score`. eq. (5) is the `scientific_plus_diversity` case. The alternatives are `scientific_only` (drops `div`); `depth_penalized`, adding $-\lambda \text{depth}(n)$ on top of `div(n)` with $\lambda = \text{depth_penalty_lambda}$; and `ucb_like`, adding $c_{\text{ucb}} \sqrt{\log N / (e(n) + 1)}$ on top of `div(n)`, where $c_{\text{ucb}} = \text{ucb_c}$ and $N = \sum_n e(n') + |\mathcal{F}|$.

Axis set. Selected by `evaluator.axis_mode`. `dynamic` (default) builds the axis set from the generic 5-axis floor plus the active rubric and `idea.json` plan keywords. `legacy` pins to the 5 canonical axes. `custom` feeds an explicit $\{(\text{name}, \text{description}, w)\}$ list verbatim through to the judge LLM.

Selection prompts. Set via `bfts.select_prompt` and `bfts.expand_select_prompt`: a pair of `FilesystemPromptLoader` keys lets the user swap in alternative templates for the two `selectLLM` calls in algorithm 1 without editing source code. The selection template must accept the placeholders `experiment_goal`, `memory_context`, and `candidates`; the expansion-target template accepts `experiment_goal` and `candidates`.

3.4 Expansion (one child per call)

The `expand` function generates *at most one* new child per invocation (no pre-batching). Callers wanting more children for the same parent call `expand` repeatedly, threading the previously created children through

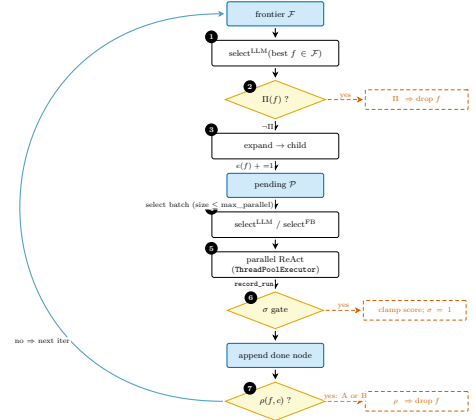


Figure 7: State-machine view of one outer BFTS iteration as a single top-to-bottom flowchart. Steps 1–3 are the inner expansion sub-loop; steps 4–7 are the outer parallel-run loop. Yellow diamonds are the predicate guards (Π, σ, ρ); red dashed boxes on the right are the side-effects each predicate triggers (frontier drop / score clamp / retire). Auxiliary state not drawn here: the per-parent expansion counter $e(\cdot)$ is bumped at step 3 and consulted by ρ at step 7, and the label history H (window W) is appended at step 5 via `record_run` and consulted at step 4 to compute the diversity bonus `div(·)`. Compare with fig. 4, which shows the same loop as a finer-grained control-flow spine with the LLM prompts drawn in.

the existing_children argument so the LLM can avoid proposing duplicate directions.

The label of each new child is decided by the LLM rather than by a hardcoded template; the expansion prompt, reproduced verbatim as section B.1, is fed:

- the parent’s status (succeeded / failed/no-real-data) and `\_scientific\_score`;
- the parent’s structured `node_report` (`delta_vs_parent` / `concerns` / `hints`, produced by the node-report builder) when present, so the planner can target specific weaknesses;
- sibling scores at the same depth, and ancestor scores from root to parent;
- tree-wide diversity metrics (unique labels seen so far, depth distribution); and
- the labels of children already spawned at this parent, so duplicates are not proposed.

A child is created in state open, handed to the agent loop for execution (section 3.7), and transitions to done when all axes are populated or to failed if the agent loop raises.

3.5 Pruning and termination

`should_prune` implements the pruning predicate Π (eq. (1)): total-node cap, depth cap, and the sterile flag. The depth check activates the `max_depth` config field. The frontier retire predicate ρ (eq. (3)) is applied after each node completes: Rule A retires a parent

Algorithm 1 BFTS run-loop (one outer iteration).

Input: state $S = (\mathcal{F}, \mathcal{P}, e, H)$; config $(B_N, B_D, E_{\max})$ and worker cap $\text{max_parallel} = \min(\text{max_parallel_nodes}, 4)$

Output: updated state S'

```

1:  $\triangleright$  — inner expansion sub-loop —
2: while  $\mathcal{F} \neq \emptyset \wedge |\mathcal{P}| < \text{max\_parallel} \wedge |\mathcal{N}| < B_N$  do
3:    $f \leftarrow \text{select}_{\text{LLM}}(\mathcal{F})$   $\triangleright$  select_best_to_expand
4:   if  $\Pi(f; |\mathcal{N}|)$  then
5:      $\mathcal{F} \leftarrow \mathcal{F} \setminus \{f\}$ ; continue
6:   end if
7:    $c \leftarrow \text{expand}(f, \text{depth\_note}, \text{budget\_note})$ 
8:    $\mathcal{P} \leftarrow \mathcal{P} \cup \{c\}$ ;  $e(f) += 1$ 
9: end while
10:  $\triangleright$  — outer parallel-execution batch —
11:  $B \leftarrow \emptyset$ 
12: while  $|B| < \min(\text{max\_parallel}, |\mathcal{P}|)$  do
13:    $n \leftarrow \text{select}_{\text{LLM}}(\mathcal{P})$   $\triangleright$  select_next_node: one node
14:    $\mathcal{P} \leftarrow \mathcal{P} \setminus \{n\}$ ;  $B \leftarrow B \cup \{n\}$ 
15: end while
16: for all  $n \in B$  in parallel do
17:    $n' \leftarrow \text{ReAct}(n)$   $\triangleright$  AgentLoop.run, may fail
18:    $H \leftarrow (H \parallel \text{label}(n'))[-W:]$   $\triangleright$  record_run
19:    $\mathcal{F} \leftarrow \mathcal{F} \cup \{n'\}$ 
20:   for all  $p \in \mathcal{F} : p = \text{pa}(n')$  do  $\triangleright$  Rule A
21:     if  $r(n') > r(p)$  then
22:        $\mathcal{F} \leftarrow \mathcal{F} \setminus \{p\}$ 
23:     end if
24:   end for
25:   for all  $p \in \mathcal{F}$  do  $\triangleright$  Rule B
26:     if  $e(p) \geq E_{\max}$  then
27:        $\mathcal{F} \leftarrow \mathcal{F} \setminus \{p\}$ 
28:     end if
29:   end for
30: end for
31: return  $S' = (\mathcal{F}, \mathcal{P}, e, H)$ 

```

f once a child’s reward $r(\cdot)$ surpasses it, and Rule B retires f once it has been expanded $E_{\max}$ times. ρ does not delete the node — its history stays on $\mathcal{T}$ — but the frontier pool shrinks so the search spreads rather than mining the same parent indefinitely.

The loop terminates when any of:

- the global `max_total_nodes` cap is hit;
- *every* frontier node fails Π (depth cap, sterile, or budget exhausted) and the pending pool is empty;
- the per-call step / cost budgets enforced by the cost tracker run out;
- the *stagnation detector* `detect_stagnation` observes that the running max of `_scientific_score` has not improved over a sliding window;
- a `LineageDecision` from the lineage-decision module explicitly halts the branch (see section B.4).

Which of these fires most often is an empirical question that section 5 is meant to answer once a

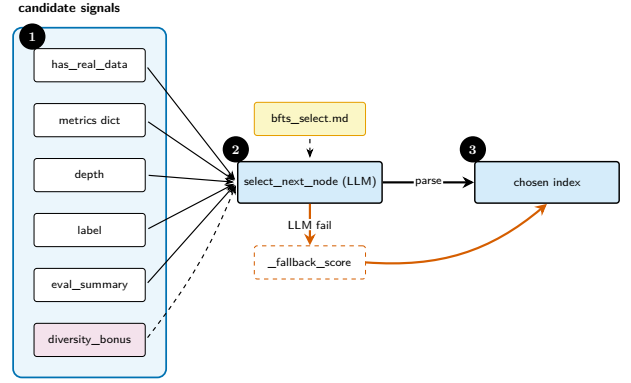


Figure 8: Selection signals fed to the LLM that picks the next node to operate on. Inputs are concatenated into a structured candidate description; the soft `diversity_bonus` is a tiebreaker, not a primary signal.

frozen checkpoint is in place; the present report does not claim a distribution.

3.6 Lineage and reproducibility

A lineage is the chain of ancestors of the best node, written into the checkpoint as the `lineage` key in `tree.json`. The `LineageState` dataclass captures the running statistics that BFTS uses to decide whether to continue, and the cross-checkpoint walker `walk_ancestor_ckpts` provides parent pointers across runs. Idea pools are inherited via `get_idea_pool_for_ckpt`, and a ranked root-idea selection is recorded by the `RootChoice` dataclass so the choice is auditable post hoc (prompt: section B.5).

Reproducibility hinges on three invariants. (i) Every randomised operation seeds itself from the node identifier. (ii) Every tool call records its arguments and outputs into the node’s working tree, never the parent’s — this is enforced by the MCP layer’s `cow_node_id` check. (iii) The cost ledger attaches a per-node dollar amount at `CostTracker.init`, so the budget check is deterministic given the same prompts.

3.7 ReAct driver

Each node’s expansion runs a ReAct-style agent loop in the `AgentLoop` class. The maximum step count is `MAX_REACT_STEPS = 80`; it can be overridden per-instance via the `max_react_steps` keyword. A separate guard `MIN_TOOL_CALLS = 2` prevents trivially short trajectories. The agent’s system prompt is reproduced verbatim as section B.6.

The set of tools available to the agent is filtered per phase by the tool manager; for example, the exploration phase masks the paper-writing tools so that BFTS expansion cannot short-circuit into a draft. A small set of MCP tools is reserved for `ari-core` itself (memory CoW operations) and is hidden from the LLM, ensuring that the model cannot set an arbitrary node id and bypass the memory skill’s copy-on-write check.

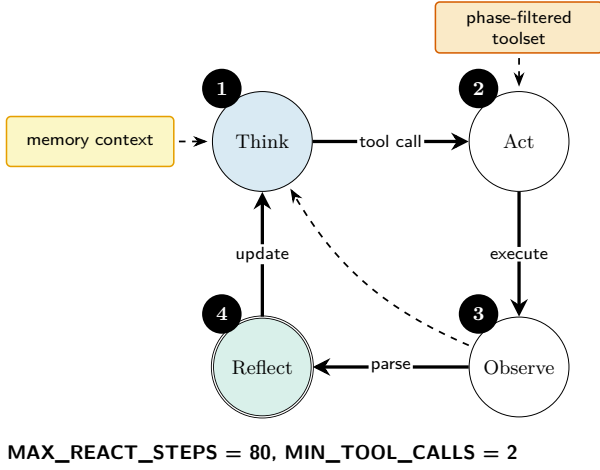


Figure 9: The ReAct loop. Solid edges are the main cycle; the dashed back edge represents an early-exit when the observation already closes the open question.

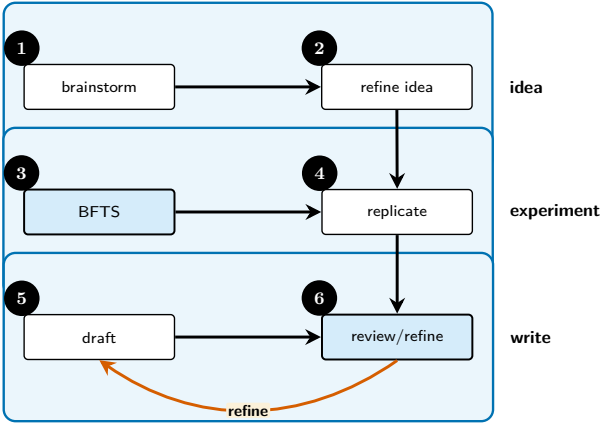


Figure 10: Paper-generation swim lane: idea, experiment, write. Each lane corresponds to a phase; the back edge from review to draft is the only non-monotonic transition (section 4.3).

4 Paper Generation Pipeline

4.1 Pipeline overview

Once exploration terminates, the surviving lineage is handed to the paper-generation pipeline, the second tree of activity in ARI (fig. 10). A *write* phase compiles the lineage into a draft, and a *review* phase scores that draft against a rubric; the two are coupled by a refine loop (section 4.3). The pipeline assembles, from the per-node artefacts left by exploration, the evidence the writer is allowed to draw on, and persists three outputs: the $\text{\LaTeX}$ draft, the per-leaf review with its rationale, and the rubric instance used to score the run (kept because the rubric may be generated per paper rather than fixed in advance).

4.2 Rubric formalism

We model the *rubric* R as a finite tree whose leaves carry tuples $(c_i, w_i, \text{judge}_i)$. Each leaf i has a category descriptor c_i , a non-negative weight w_i with

$\sum_i w_i = 1$, and a *judge function* $\text{judge}_i : P \rightarrow [0, 1]$ that returns a graded score. The paper score is the convex combination

$$\text{score}(P) = \sum_i w_i \text{judge}_i(P). \quad (6)$$

This tree-structured rubric is shared with PaperBench, which ARI reuses through a vendored package (section 4.4) so that its grade and aggregation semantics track upstream verbatim. Two judge families coexist: an LLM-backed judge for qualitative leaves, and a deterministic judge for leaves with a binary or numeric verifier (e.g. “is the cost reported in absolute dollars?”). Both satisfy the same *Evaluator* interface, so the search engine (section 3) and the paper pipeline reduce axes to a scalar through one seam.

A rubric is generated in one of two *modes*. In *agent-benchmark* mode the tree decomposes the paper by scientific contribution and each leaf asks whether a candidate submission’s executed output reproduces the paper — the original PaperBench framing. In *paper-audit* mode the tree’s top-level children are a fixed set of reproducibility axes and each leaf grades the descriptive completeness of the paper text itself — a framing inversion aimed at HPC reproducibility audits, where the question is not “can an agent reproduce this paper?” but “does this paper describe enough to be reproducible?”. Venue knowledge (which axes, how they are phrased) is supplied by a per-venue template rather than baked into the engine, so ARI core stays domain-agnostic (P4) and a new venue is a data change, not a code change.

Lifting the paper-audit score out of a degenerate regime — where an empty submission drives every submission-oriented leaf to “no” — required rephrasing the leaf questions to interrogate the *paper* rather than a (missing) submission, feeding the judge the paper’s figures alongside its text so vision-capable models can use them, and admitting optional artifact-description and artifact-evaluation appendices as extra input. These adjustments live entirely in ARI’s wrapper layer; the vendored judge is untouched, which keeps scores comparable across upstream versions. We deliberately do *not* synthesise a reproduction log from the paper’s own reported numbers: doing so would short-circuit the executed-submission stage and inflate the score in a way not comparable to leaderboard runs.

4.3 Review/refine loop

The refine cycle iteratively rewrites P_t to P_{t+1} under the review feedback $J(P_t)$:

$$P_{t+1} = \text{Refine}(P_t, J(P_t)). \quad (7)$$

The cycle stops when the reviewer accepts the draft — modelled as the *convergence criterion* $\text{score}(P_{t+1}) - \text{score}(P_t) < \epsilon$ — or when a small per-section revision cap is reached. Which branch fires more often is an empirical question section 5 is meant to answer once

a frozen checkpoint exists; we make no quantitative claim here.

The refine step is intentionally non-monotonic per axis: fixing a clarity issue may slightly degrade a numeric axis. This is the source of the back edge in fig. 10, and it is why the convergence criterion is defined on the aggregate score rather than per-axis — together with a per-axis floor (section 4.6) that prevents the aggregate from hiding a collapse on any single axis.

4.4 PaperBench integration

The replication path is *not* a re-implementation of PaperBench: the upstream package is vendored unchanged so that ARI tracks its task and rubric semantics verbatim, and ARI contributes only a thin wrapper around it. The wrapper exposes the protocol-faithful loop as three composable stages with a shared vocabulary (fig. 11):

1. *rollout* — a replication agent works from the paper towards a submission (code, and optionally a reproduction script);
2. *reproduce* — the submission is executed in an isolated container, on the local machine or dispatched to a cluster;
3. *grade* — the vendored judge scores the submission against the rubric tree.

Two design choices keep this faithful to the benchmark. First, grading is left to the vendored judge; ARI’s adapters only reshape its output and aggregate repeated runs into a single score, so upgrading PaperBench is a vendor swap with no change to how leaf grades are combined — the same envelope feeds eq. (6), with $\text{judge}_i \in \{0, 1\}$ for binary leaves. Second, the wrapper *fails loud*: when a benchmark-fair run cannot be served (no container runtime, no scheduler, no GPU resource registration) it raises rather than silently degrading to a weaker host, because a silent downgrade would make the resulting score incomparable to a leaderboard entry. Legacy silent-fallback behaviour is available only behind an explicit opt-in.

A second concern is keeping the replication agent honest about what the host actually provides. Left to the upstream prompt, an agent tends to scaffold a reproduction script that calls binaries the host lacks, and to default to Python regardless of the paper’s real language. ARI counters both by priming the agent to probe the environment before it scaffolds, and by injecting a per-host notes block — generated from a live probe of the available toolchain, GPUs, and network reachability — so that what the agent reads matches what the host can do.

4.5 Domain breadth: the execution-profile contract

PaperBench upstream assumes a single container with one GPU. To cover papers whose methodology is fun-

damentally parallel — multi-rank MPI kernels, multi-node training, cluster-specific module environments — ARI extends the rubric with an optional *execution profile* (fig. 12). The profile is domain-agnostic: it names a *parallel-execution shape* (single-CPU, single- or multi-GPU, MPI, MPI+GPU) together with the paper’s scale envelope and any scheduler hints, rather than anything about the paper’s scientific field. When a paper is single-machine the profile is omitted and the pipeline degenerates to the original single-node invocation: the contract is additive, not breaking.

The profile feeds two consumers (fig. 12). It is appended to the agent’s prompt as a compact “compute-node execution conventions” block — shared-filesystem discipline, a preference for the scheduler’s launcher over a bare MPI launcher, environment activation, and wall-clock wrapping — and it parameterises the cluster dispatch (node, GPU, memory, and placement requests). Because clusters disagree on which scheduler flags they accept, the dispatcher probes the local scheduler at run time and drops flags it does not support, so the same rubric dispatches faithfully on a multi-GPU cluster, on a sandbox partition without GPU registration, or on a single workstation. A regression guard enforces the inverse: ARI running inside an unrelated cluster allocation must never leak cluster- or MPI-specific prose into the prompt for a non-HPC paper. Only this wrapper layer changes; the vendored benchmark and ARI core are untouched across the extension.

4.6 Failure modes and guardrails

Three failure modes are common enough to be named, each with a structural — not merely advisory — mitigation:

- **LLM-only support:** the paper asserts a claim that no node in the lineage substantiates. The mitigation constrains the writer to a pre-selected set of lineage artefacts, so every numerical or causal claim is traceable to a node that produced it; the draft cannot cite evidence that was not generated.
- **Citation hallucination:** the paper cites a non-existent work. The mitigation is structural — the writer may not invent citation keys, and every bibliography entry is verified against an external index before it is admitted (section 6).
- **Refine drift:** the aggregate score climbs while the paper loses coherence. The mitigation is a per-axis floor: if any axis falls below a threshold the refine is aborted, so a gain on one axis cannot mask a collapse on another.

The first guard also bounds context: the writer is given only the bytes of the selected lineage artefacts, under a size budget, so the draft stays within the model’s context window even for long runs.

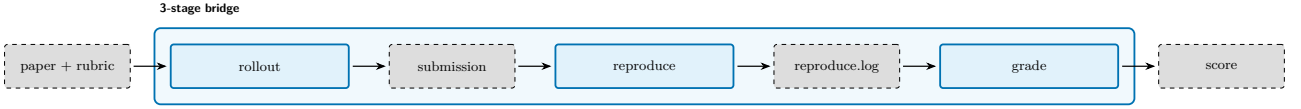


Figure 11: The PaperBench replication path as three composable stages. The agent *rollout* turns the paper and its rubric into a submission; *reproduce* executes that submission in an isolated sandbox, capturing `reproduce.log` and its outputs; *grade* scores the submission against the rubric tree. ARI’s three-stage bridge orchestrates the stages and aggregates repeated runs into a single score.

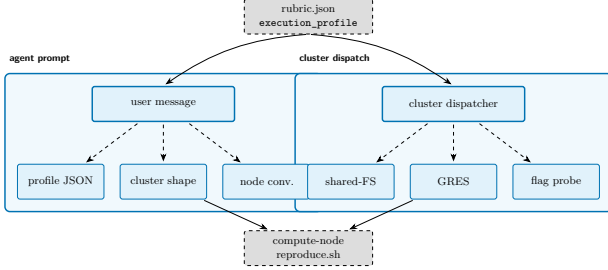


Figure 12: The execution-profile contract threads from a single rubric artefact into two consumers: the replication agent’s prompt (left) and the cluster dispatcher (right). Runtime probes adapt the dispatched resource request to the local scheduler, so one rubric runs faithfully on heterogeneous clusters without being modified.

5 Preliminary Observations

The observations below are what the running system establishes at this stage. They are preliminary and, for the end-to-end reproduction case, single-run rather than a controlled benchmark; a panel-scale quantitative study is left to future work. Quantitative figures elsewhere in this report are illustrative, generated from fixed-seed sample data so that the document build is reproducible.

Three properties hold directly by construction of the system. Cost accounting is exact: a dollar cost is attached to every model invocation and accumulated, per run, into the checkpoint’s results record. The exploration loop terminates through the stagnation detector (section 3.5) rather than only the step or cost budgets ($T_{\max}$ / $C_{\max}$). And the execution-profile pathway behaves as specified (section 4.5): rubric generation emits a multi-rank (MPI) execution profile, with the paper’s reported scale envelope, for an HPC paper and omits it for a single-machine paper, while a multi-flag cluster dispatch is accepted by the local scheduler once the runtime probes drop the flags it does not support.

End to end, the v0.8.0 pipeline was exercised on a lossy-compression target, a GPU error-bounded scientific compressor. The degree of reproduction achieved on this single target was partial but substantive. The agent fetched a real slice of the paper’s cited simulation dataset rather than substituting synthetic data; compiled native CUDA interpolation kernels and ran them on the GPU; implemented Lorenzo and multi-level interpolation predictors with per-level autotuning; and swept a range of error bounds, recovering reconstruction qualities from roughly 50 to 90 dB PSNR with

the corresponding compression ratios. Against the fine-grained rubric the run cleared on the order of a tenth of the weighted leaves, for two reasons rooted in the agent’s behaviour: the graded metric came from a CPU predictor — the GPU kernels were built and executed, but the agent omitted their predicted quality from its reported results — and the agent terminated voluntarily after a small fraction of its time budget, leaving the remaining datasets and the proprietary lossless stage unattempted.

A controlled evaluation — median run cost, per-seed exploration curves, and per-category rubric distributions over a frozen checkpoint, together with full-rollout replication scores across a panel of contrasting target papers — is left to future work.

6 Related Work

We position ARI against four threads of prior art.

6.1 Tree search for code

AIDE [2] formalises ML-engineering as a tree search over plan / code / debug states; AlphaCode [3] is the canonical reference for large-scale generate-and-filter on competitive programming; the MLE-bench evaluator [4] is now the standard for measuring ML-engineering agents. ARI inherits the BFTS skeleton from AIDE but adds (i) an LLM-driven candidate selector (section 3.3), which avoids fixing a closed-form scalar utility and lets the selection prompt weigh `has_real_data`, full `metrics`, depth, and a soft `diversity_bonus` jointly, and (ii) the cross-run lineage (section 3.6) which neither AIDE nor MLE-bench formalises.

6.2 Autonomous research agents

The AI Scientist family [5, 6] closes a similar loop end-to-end (idea → experiment → paper) but does not expose the rubric and refine cycle as a first-class object. VirSci [7] simulates several scientist personas and uses their consensus to score outputs; Agent Laboratory [8] treats the agent as a collaborator rather than a driver. ARI differs in two operational respects: **everything lives in a checkpoint** (P1 / P4), and the rubric is the contract between explore and write.

6.3 Paper evaluation

PaperBench [9] is the closest analogue to our **paper-re** integration; it pioneered the rubric-as-tree formalism. We adopt the same per-leaf grade and aggregate weight (eq. (6)) but extend it with a per-axis floor used as a refine guard (section 4.6).

6.4 Foundations

The agent loop follows the ReAct pattern [1]; the refine cycle is descended from Self-Refine [10] and Reflexion [11]; and the choice to give the LLM a step-by-step scratchpad in *Think* is descended from Chain-of-Thought [12]. None of these references propose a checkpoint-scoped state; that is ARI’s contribution.

An earlier draft cited a separate ML-research-agent benchmark, but we removed that reference once it became clear that none of the candidate sources met our verification rule (section 4.6); we keep the related-work coverage to entries that can be S2-verified.

7 Discussion and Limitations

7.1 Determinism vs novelty

P2, the determinism principle, is the binding constraint of the system. Strictly enforcing P2 forces *ari-skill-memory* to declare “no LLM calls” — it must score its retrievals by a deterministic keyword function rather than an embedding model whose output depends on a remote service. The trade-off is that retrieval quality is bounded by what keyword scoring can recover.

This is not an accident: every reproducibility win comes at the cost of a closed door for non-determinism. We accept the cost because the checkpoint, not the run, is the contract with downstream users.

7.2 Scaling limits

The single-machine assumption underlying the BFTS run-loop is the biggest scaling constraint. A run that wants to expand $b > 10$ in parallel needs a different concurrency layer; today a single `ThreadPoolExecutor` in the BFTS CLI loop is the sole worker pool. A multi-host expansion would require extending the lineage discipline (P4) to handle merges, not just ancestor walks.

A separate constraint is the cost of the LLM grader. Empirically the grader is one of the dominant line items in a run, alongside the explorer; whether it is the second-largest in absolute dollars is something section 5 is meant to settle once a frozen checkpoint exists. Caching grader outputs by (paper-hash, rubric-hash) would help; we have not yet shipped this.

A Notation Table

This appendix collects every symbol used in the body of the report, with its meaning.

Symbol	Meaning
$\mathcal{N}$	set of nodes in the search tree
$\mathcal{T}$	search tree $(\mathcal{N}, E)$
$n \in \mathcal{N}$	individual node
$\text{ch}(n)$	children of n
$\text{pa}(n)$	parent of n
$\text{depth}(n)$	depth of n
$s(n)$	node state $\in \{\text{open}, \text{eval}, \text{done}, \text{failed}\}$
$r(n)$	scalar reward of n (its <code>_scientific_score</code> $\in [0, 1]$)
$\mathbf{e}(n)$	per-axis evaluation vector
ϕ	axis $\rightarrow$ scalar aggregator
$\text{div}(n)$	diversity bonus (eq. (4))
$\text{fb}(n)$	deterministic fallback ranking (eq. (5))
$T_{\max}, C_{\max}$	step / cost budgets
R	rubric tree
c_i	rubric leaf i 's category
w_i	rubric leaf i 's weight
judge_i	rubric leaf i 's judge
P	paper draft
$\text{score}(P)$	paper aggregate score (eq. (6))
Refine	refine operator (eq. (7))
ckpt	checkpoint directory
$\text{lin}(n)$	lineage chain ending at n

B Runtime LLM Prompts

This appendix lists the LLM prompts that ARI uses at runtime, reproduced verbatim from `ari-core/ari/prompts/` at v0.8.0. Each listing is byte-for-byte identical to the source on disk. Because the exact bytes are what reach the model, the prompts are reproduced as-is in every language version of this report and are not translated.

Orchestrator

B.1 BFTS expansion

```
You are expanding a BFTS research tree node.

{goal_line}Parent node id={parent_id_short}, depth={parent_depth}, status={parent_status}
{depth_note}{budget_note}Parent metrics: {parent_metrics_json}
Parent summary: {parent_summary}
{sci_note}{idea_block}{parent_report_block}
{siblings_block}{ancestors_block}{existing_block}{diversity_block}Propose exactly ONE child research direction
that is the most scientifically valuable next step. For the "label" field, strongly prefer one of the canonical
labels [draft, improve, debug, ablation, validation]; only invent a custom label (e.g. "replication",
"generalization") when none of the five fits meaningfully - the custom string will be preserved as raw_label.
Base your choice on the experimental context above, not on a fixed template.

Label selection guidance:
- 'debug': parent FAILED or has no real data - diagnose and fix it.
- 'improve': parent succeeded and you want to push its metrics higher by tuning parameters, flags, or
algorithms.
- 'ablation': isolate which component drives the parent's gains by removing or varying ONE component. State
explicitly what is removed/varied and what delta vs. the parent metrics you expect.
- 'validation': rigorously verify the parent's claims (different seeds, edge cases, stress tests,
expected-degradation checks).
- 'draft': start a fresh implementation from scratch to introduce a fundamentally NEW perspective (use this
instead of inventing a new label like 'replication' or 'generalization').

Reply ONLY with a JSON array containing exactly one element: [{"label": "<one of:
draft|improve|debug|ablation|validation>", "direction": "..."}]
Example: [{"label": "validation", "direction": "<one-sentence plan>"}]
```

B.2 BFTS expansion-time selection

You are selecting which completed research node to expand next in a BFTS tree.

Experiment goal: {experiment_goal}

Completed nodes awaiting expansion:
{candidates}

Select the single most promising node to expand. Prefer nodes with high scientific_score, strong metrics, and unexplored directions. Avoid nodes that have already been retried many times or are at excessive depth. Reply with ONLY the index number (0-based).

B.3 BFTS selection

You are selecting the most promising node to explore next in a research tree.

Experiment goal: {experiment_goal}

Relevant past memories:
{memory_context}

Candidates:
{candidates}

Selection criteria:

- Nodes with has_real_data=True and strong metrics are high-value
- Consider all metrics holistically (multi-objective)
- Deeper nodes with excellent results are worth continuing
- A small diversity_bonus is awarded to underrepresented exploration types; treat it as a soft tiebreaker, not a primary signal

Reply with ONLY the index number (0-based) of the best node.

B.4 Lineage continuation decision

You are a research orchestrator. Given the current state of an exploratory research run, decide the next lineage-level action. Possible actions:

- continue - current idea is still productive; keep exploring
- switch_to_idea - current idea has stagnated; switch to an alternative
- fanout - current idea succeeded; explore an alternative in parallel
- terminate - research thread is exhausted; stop

Choose the LEAST disruptive action that fits the evidence. Prefer continue unless there is a concrete reason to escalate. When choosing switch_to_idea or fanout, set target_idea_index to a value that appears in the alternatives pool. When choosing switch_to_idea, set disable_generate_ideas=true (the child should run with the chosen idea pinned, not regenerate). For fanout you may set it false so the child can also explore additional novel directions.

Reply ONLY with JSON:

{"action":str,"target_idea_index":int|null,"disable_generate_ideas":bool,"rationale":str}. No markdown.

B.5 Root idea selector

You are a research orchestrator picking the ROOT idea for a run from a VirSci-generated pool. VirSci has already scored each idea by novelty/feasibility/clarity, but you have additional context (venue rubric, ancestor research thread, run notes). Your job is to pick the idea most likely to produce a strong submission for this venue, considering all signals.

Rules:

- chosen_index MUST be an integer that exists in the pool.
- Default to VirSci's choice (index 0) UNLESS another idea is clearly stronger given the additional context.
- One sentence rationale describing your reasoning.

Reply ONLY in JSON: {"chosen_index": int, "rationale": str}. No markdown.

Agent loop

B.6 ReAct agent system prompt

You are a research agent. You MUST use tools to execute experiments. Do NOT write plans or text descriptions - call a tool immediately.

AVAILABLE TOOLS:


```
{tool_desc}
```

RULES:

- Your FIRST action must be a tool call. Never output a text plan.
- If `make_metric_spec` tool is available and this is a new experiment (not a continuation), call it early to self-determine evaluation criteria.
- NEVER fabricate numeric values - only report values from actual tool outputs
- AFTER your final measurement run, when `emit_results` is available, call it once to record a typed split between INPUT parameters (matrix size, thread count, seeds - knobs you ran on) and MEASUREMENTS (throughput, accuracy, latency - what you measured). This lets downstream stages distinguish "what we ran on" from "what we measured" so a best-of reduction never picks an input size as the result. Do NOT include input parameters in `measurements` and do NOT include measured outputs in `params`.
- When all experiments are done, return JSON: `{"status": "success", "metrics": {...}, "summary": "..."}{extra}`
- Do NOT call `gap_analysis` or `generate_hypothesis`
- Ensure your experiment is reproducible: capture whatever information would be needed for an independent researcher to reproduce your results and verify your findings

Evaluator

B.7 Peer-review evaluator

You are a research data extractor AND a scientific peer reviewer.

Analyze the experiment artifacts and return a JSON with:

```
has_real_data: bool (true only if numeric measurements appear in artifacts)
params: dict of INPUT/configuration values (NOT measurements). Examples: matrix size, thread count, seed.
measurements: dict of MEASURED quantities (the experiment's output). Examples: GFLOP_per_s, accuracy,
latency.
metrics: dict - flat union of params and measurements (back-compat).
reason: str (one sentence describing what was measured)
{axes_block}
comparison_found: bool (true if results involve comparison with existing approaches)
```

When scoring `claim_implementation_alignment`, look for concrete preconditions or contracts the plan / model states (e.g. 'eliminates RFO traffic', 'preserves equivariance', 'requires sorted input') and cross-reference them against the artifacts. Score low when the implementation contradicts a stated assumption, even if the kernel runs without errors.

Return ONLY valid JSON, no markdown fences.

B.8 Metric extraction

You are a research data extractor AND a scientific peer reviewer.

Analyze the experiment artifacts and return a JSON with:

```
has_real_data: bool (true only if numeric measurements appear in artifacts)
params: dict of INPUT/configuration values the experiment ran on. These are knobs, not outcomes. Examples:
matrix dimensions (M, K, nnz), thread count, batch size, ISA flags, seeds. They are NOT candidates for a
best-of reduction.
measurements: dict of MEASURED quantities - what a peer reviewer would treat as the experiment's result.
Examples: GFLOP_per_s, GB_per_s, latency_s, accuracy. ARE candidates for best-of.
metrics: dict - for back-compat, the union of params and measurements as a single flat dict (e.g.
{"GFLOP_per_s": 754.8, "M": 120000}). Downstream consumers may read either the typed split above or this
flat view. NEVER include the same key in both views with different values.
reason: str (one sentence describing what was measured)
axis_scores: dict with EXACTLY these five keys, each a float in 0.0-1.0:
- measurement_validity: are numeric measurements present and methodologically sound?
- comparative_rigor: are results compared against baselines or prior work?
- novelty: does the work advance beyond existing approaches?
- reproducibility: is there enough detail for someone else to reproduce the result?
- clarity_of_contribution: is the scientific claim stated clearly and specifically?
Score 0.0 on an axis when that dimension is absent or fatally weak. Score toward 1.0 as the dimension
approaches publishable quality. These axes are combined via weighted harmonic mean, so a low score on
ANY axis drags the overall score down - differentiate axes deliberately.
axis_rationales: dict with the same five keys; each value is one sentence explaining the score for that
axis.
comparison_found: bool (true if results involve comparison with existing approaches)
Return ONLY valid JSON, no markdown fences.
```

Pipeline

B.9 Keyword librarian

You are a research librarian. Given experiment summaries, produce a SHORT broad academic search query (3-6 words) for Semantic Scholar. Use GENERAL terms (e.g. 'algorithm optimization', not 'custom 4-stage pipelined batch-norm fused kernel'). Avoid domain-specific jargon, acronyms, or overly narrow terms. Return ONLY the query string, no explanation.

Wizard (Viz)

B.10 Wizard goal chat

You are an assistant helping a researcher set up an ARI experiment. ARI automatically writes, runs, benchmarks code, then produces a paper. Ask focused questions ONE AT A TIME to understand: (1) what to optimize or investigate, (2) how to measure success (metric name and direction), (3) platform/hardware constraints, (4) any baseline to compare against. Be concise. After 3-5 exchanges and sufficient info, output EXACTLY: ---READY--- followed by a complete experiment.md with sections: ## Research Goal, ## Evaluation Metric, ## Constraints. Do NOT output the MD before getting at least 2 user responses.

B.11 Wizard config generator

You are helping a researcher set up an automated experiment. Convert the following research goal into a concise experiment.md file with a ## Research Goal section. Keep it to 3-5 sentences maximum. Do not add code or technical details. Research goal: {goal}

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